Vishnu Institute of Technology(A) :: Bhimavaram

Databricks Foundations Training Program

For Capgemini-Selected Engineering Graduates

Vishnu Institute of Technology, Bhimavaram, Andhra Pradesh

Course Overview

Version 2.0 | March 2026

1. Program Overview

Program Title	Databricks Foundations for Data Engineering
Institution	Vishnu Institute of Technology, Bhimavaram, Andhra Pradesh
Target Audience	60 Final-Year Engineering Graduates (Capgemini-Selected)
Expected Roles	Junior Data Engineer / Databricks Support / Data Platform Associate
Duration	3 weeks with 15 Working Days (Monday to Friday, Saturday as buffer day) - Total 90 Hours, **starting on 6th April, 2026**.
Daily Schedule	6 Hours per Day
Delivery Mode	Instructor-Led, Hands-On Training (In-Person)

2. Program Objective

This program is designed to build strong foundational skills in Databricks, PySpark, SQL and Delta Lake for students who have been selected by Capgemini for junior Data Engineering and Databricks support roles.

By the end of this program, students will be able to:

- Navigate the Databricks workspace confidently - notebooks, clusters, DBFS and catalogs
- Write solid SQL for data engineering tasks - joins, window functions, CTEs, deduplication and incremental logic
- Write production-grade PySpark pipelines and translate SQL logic into PySpark DataFrame operations- select, filter, join, aggregate and handle nulls
- Understand and work with Delta Lake - ACID tables, MERGE operations, schema evolution and time travel
- Build a complete Medallion pipeline (Bronze → Silver → Gold) with basic data quality checks
- Create and schedule Databricks Jobs and Workflows with parameters and task dependencies
- Read logs, interpret error messages and diagnose common pipeline failures
- Understand Unity Catalog basics, Git-based collaboration and cloud storage fundamentals
- Complete a team-based capstone project demonstrating end-to-end pipeline development

3. Program Philosophy: What This Is (and Isn't)

This Program IS	This Program is NOT
A strong foundation in Databricks, SQL, PySpark and Delta Lake	Going to cover other technologies like Snowflake, DBT, etc.
Hands-on, lab-intensive learning with strong instructor-led sessions	A theory-heavy lecture series with minimal practice
Focused on patterns juniors will actually encounter on the job	An advanced engineering bootcamp covering DevOps, MLOps or infra
Building debugging confidence and problem-solving mindset	Deep streaming, Kafka, advanced Spark internals or ML workloads
Designed to reduce onboarding ramp-up time at Capgemini and make students as close to being production ready as a Databrick's Data Engineer.	Replacement of Experience that comes as a practitioner with time.

4. Target Skill Levels by End of Program

Skill Area	Entry Level (Before)	Target Level (After 90 Hours)
Data Warehouse & ETL	No exposure	Understand what they are learning and fundamentals properly.
Git / Collaboration	Minimal	Can clone repos, commit changes and follow notebook standards
SQL	Basic/academic SQL	Solid working SQL: joins, windows, CTEs, dedup patterns, Databucketing
PySpark / DataFrames	No exposure (Python basics assumed)	Can write DataFrame transformations, joins, aggregations
Cloud Storage Awareness	No exposure	Conceptual understanding of ADLS/S3 and external storage
Delta Lake	No exposure	Understand ACID, MERGE, time travel, schema evolution
Medallion Architecture	No exposure	Can design and build Bronze → Silver → Gold pipelines
Jobs & Workflows	No exposure	Can create, schedule and parameterize multi-task jobs
Debugging / Troubleshooting	Minimal	Can read logs, identify common failures and apply fixes
Data Governance	No exposure	Awareness of Unity Catalog, namespaces, basic permissions

5. Daily Session Structure

The program runs for 15 working days, covering 6 hours of training per day for a total of 90 hours. Each day's 6-hour session is structured as a balanced mix of instructor-led teaching and practical application:

Component	Approx. Hours/Day	Description
Instructor-Led Teaching & Demonstrations	~2 hours	Detailed concept explanations, live coding walkthroughs, architecture discussions and real-world scenario demonstrations by the trainer
Guided Hands-On Labs	~3 hours	Structured practical exercises where students work on labs with instructor guidance and support available throughout
Daily Quiz, Review, Troubleshooting & Q&A	~1 hour	Short daily quiz or coding challenge to reinforce learning, followed by solution review, discussion of common mistakes, troubleshooting and open Q&A

5A. Daily Hands-On Evaluation & Reinforcement

To ensure strong practical learning outcomes, each day will include:

- A **graded hands-on lab submission**
- A **short coding evaluation (SQL or PySpark)**
- Review of common mistakes and debugging discussion

Students will be expected to:

- Write code independently
- Avoid hardcoded solutions
- Demonstrate correct reasoning and approach

Daily evaluations will help identify gaps early and ensure consistent progress.

6. Course Agenda: 3 Week Breakdown

0. Pre-Training Mandatory Preparation (1 Week Prior – current week)

To ensure all students are adequately prepared and aligned with the expectations of the Databricks Data Engineering program, the following pre-training activities are **mandatory** and must be completed before the commencement of the main training:

1. SQL Readiness Assessment

- Students must complete a structured SQL assignment covering:
  - Joins across multiple tables
  - Aggregations and filtering
  - Basic analytical queries
- Submission will be collected through a structured evaluation form.
- Students will be evaluated on correctness, clarity, and approach.

2. Python Data Processing Assignment

- Students will solve data-processing problems using Python (without SQL).
- Focus areas:
  - Reading structured data
  - Transformations and aggregations
  - Logical problem-solving

3. Databricks Academy (Mandatory Modules)

- Students must complete:
  - Associate Data Engineering learning track (initial modules)
- Completion proof must be submitted before training begins.

4. Readiness Expectation

- Students entering the program should:
  - Be comfortable with basic SQL and Python
  - Be able to reason across multiple tables
  - Be familiar with basic data concepts

WEEK 1: FOUNDATIONS (Days 1–5, 30 Hours, Starting 6th April)

Throughout Week 1, students will work on **scenario-based datasets (banking/e-commerce domain)** involving multiple interconnected tables.

Emphasis will be placed on:

- Understanding relationships between tables
- Writing queries across 4–6 tables
- Handling real-world data issues such as duplicates, missing values and inconsistent records

The focus will be on developing **data reasoning and problem-solving ability**, not just SQL/python/pyspark syntax.

Day	Topic	Key Modules & Sub-Topics	Learning Outcome
Day 1	**Data Warehouse and ETL basics, python for Data Engineers**	Lakehouse vs Data Warehouse vs Data Lake (use-case comparison) Data formats: CSV, JSON, Parquet Why Parquet is default in industry ETL vs ELT What can go wrong in data? Python for Data Engineering (focused refresher)	Students understand why Data Engineering exists, are aware of real-world data issues, and can create a mini ETL pipeline in python.
Day 2	**SQL Foundations for Data Engineering**	SELECT, filters, CASE, casting, NULL handling Joins: INNER, LEFT, OUTER, etc., and their practical significance Aggregations, GROUP BY, HAVING Common data pitfalls: duplicates from joins, NULL behaviour	Students can write SQL and trust their results, not just produce them
Day 3	**Advanced SQL: Job-Ready Patterns**	CTEs and subquery refactoring Window functions: ROW_NUMBER, RANK, LAG/LEAD, running totals Deduplication patterns (latest record per key) SCD Type 1 & Type 2 concepts Incremental load logic (watermark approach) Data Bucketing & Partitioning	Students can write production-grade SQL, apply window functions and de-duplication patterns, understand incremental data processing, and have a foundational understanding of data partitioning and bucketing for performance optimization.
Day 4	**PySpark Foundations: DataFrames & Transformations**	Spark architecture overview: driver, executors, partitions (conceptual) DataFrame API: read, select, withColumn, filter, groupBy, join Transformations vs Actions Functions, expressions and distributed thinking SQL vs PySpark: same logic, different syntax - side-by-side comparison	Students can write PySpark transformations, understand how Spark executes code, and translate SQL logic into distributed data processing.
Day 5	**PySpark Continued + Data Ingestion Patterns**	Handling NULLs, type casting, column ops UDF awareness (when to use, when to avoid) Batch ingestion: full load vs incremental Parameterized notebooks for repeatable data loading Reading and understanding existing PySpark code	Students can build end-to-end PySpark pipelines, understand Delta Lake concepts, and write reusable, production-style ingestion notebooks.

WEEK 2: CORE DATABRICKS & DELTA LAKE (Days 6–10, 30 Hours)

Master Delta Lake, Medallion architecture, performance and job operations

Day	Topic	Key Modules & Sub-Topics	Learning Outcome
Day 6	**Delta Lake Essentials**	What is Delta Lake: ACID transactions on data lakes Delta vs Parquet - the real difference Create, Insert, Update, Delete on Delta tables MERGE INTO for upserts (critical pattern for real projects) Schema evolution and enforcement Time travel, DESCRIBE HISTORY, RESTORE	Students can create Delta tables, perform MERGE operations and use time travel for data recovery
Day 7	**Medallion Architecture End-to-End**	Bronze: raw, append-only ingestion layer Silver: cleaned, conformed, deduplicated Gold: business aggregates and serving tables Introduction to star schema (fact vs dimension tables) Data quality checks: null checks, uniqueness, range validations	Students can design and build a complete Bronze → Silver → Gold pipeline with quality checks
Day 8	**Performance Basics & Troubleshooting**	Partitions, file sizes, small files problem Data skew: what it is and how to address it Broadcast joins concept OPTIMIZE and ZORDER: table maintenance essentials Caching and explain plans basics Spark UI walkthrough: understanding jobs, stages, tasks	Students can identify common performance issues and apply basic optimisation techniques
Day 9	**Incremental Ingestion + Databricks SQL**	Auto Loader / cloudFiles: incremental file ingestion pattern Batch vs streaming ingestion concepts Checkpointing and trigger modes (conceptual) Databricks SQL Warehouses and their purpose Creating views and serving layer patterns Validation queries and basic dashboards Conceptual understanding of Databricks integration with Azure Data Lake Storage (ADLS) and data ingestion flow from cloud storage.	Students understand incremental ingestion patterns and can write SQL validation queries
Day 10	**Workflows, Jobs & Scheduling**	Jobs vs interactive notebook runs Creating multi-task workflows with dependencies Parameters, widgets and environment config Retry logic and failure notification patterns Converting medallion notebooks into scheduled pipelines	Students can create, parameterize and schedule Databricks workflows

WEEK 3: INTEGRATION & CAPSTONE (Days 11–15, 30 Hours)

Debugging mastery, governance, collaboration and capstone project

Day	Topic	Key Modules & Sub-Topics	Learning Outcome
Day 11	**Debugging Practical: Diagnosing Pipeline Failures**	Reading Spark UI and error logs at a practical level Common failure scenarios covered through intentional breakages: Permission and access errors Table/namespace resolution issues Schema mismatch and column drift NULL pointer and logic errors Memory and shuffle issues (awareness) Data quality assertion failures Students diagnose from logs and fix each scenario	Students can read error messages, identify root causes and fix common pipeline failures
Day 12	**Governance, Git & Cloud Awareness**	Unity Catalog: 3-level namespace Permissions basics and least privilege concept Databricks Repos: clone, commit, pull, basic branching Notebook collaboration: modular cells, documentation, naming conventions Cloud storage fundamentals: What is ADLS/S3, where data lives, how Databricks connects to external storage	Students understand governance basics, can use Git within Databricks and know how cloud storage works
Day 13	**Capstone Project: Kickoff & Design**	Project scenario briefing and team formation (4–5 students per team) Requirements analysis: inputs, outputs, refresh patterns Data model design: fact and dimension tables Pipeline architecture: medallion layers, quality checks, incremental logic Begin implementation: Bronze and Silver layers	Teams have a project plan, data model and working Bronze + Silver implementation started
Day 14	**Capstone Project: Build & Harden**	Complete Silver and Gold layer implementation Implement MERGE operations and quality checks Create workflow with scheduling Add logging and data quality report outputs Run end-to-end with simulated incremental batch Notebook documentation and code cleanup	Teams have a fully functional end-to-end pipeline with documentation
Day 15	**Capstone Demos, Assessment & Wrap-Up**	Team presentations: architecture, pipeline, data quality, workflow, debugging approach Peer review and feedback Final hands-on assessment: fix a broken pipeline and deliver gold output Key interview concepts: medallion architecture, Delta MERGE, debugging approach Databricks certification guidance (Associate DE exam overview) Program completion and certificates	Students can demonstrate and articulate their work; ready to enter Capgemini's internal training with confidence

7. Training Datasets

The training will use 2–3 realistic datasets that reflect the types of data processing scenarios commonly encountered in enterprise data engineering projects. These datasets will be provided in multiple file formats (CSV, JSON, XML, Parquet) to give students experience working with diverse data sources - a core skill for any data engineering role. Each dataset will have more than 10 tables, so that dataset is realistic.

Planned datasets:

Dataset	Description	Key Learning Patterns
E-Commerce Orders & Customer Data	A comprehensive retail dataset covering orders, customers, products, payments and shipments across multiple file formats	Incremental ingestion, MERGE/upsert operations, deduplication, late-arriving data handling, star schema design, data quality checks
IoT Sensor / Device Telemetry	Time-series data from simulated devices arriving as JSON and CSV streams with varying schemas and data quality issues	High-volume data processing, schema drift handling, time-series aggregations, data validation, handling malformed records
Reference / Master Data	Slowly-changing dimension data such as product catalogues, regional configurations and lookup tables in XML and CSV formats	SCD patterns, dimension table management, XML parsing, cross-format joins, data reconciliation

These datasets will be used consistently throughout the program, building complexity progressively from Day 1 through to the capstone project. No real or sensitive data will be used at any point.

Dataset Characteristics (Critical Design Principle)

All datasets used in the program will simulate real-world complexity:

- Each dataset will include **10+ related tables**
- Data will intentionally include:
  - Missing values
  - Duplicate records
  - Inconsistent or late-arriving data
- Students will be required to:
  - Understand schema relationships
  - Apply joins across multiple tables
  - Handle data quality issues

Students will not work on simplified or clean academic datasets.
The focus is on **real-world data behavior and engineering challenges**.

8. Assessment Framework

Student progress is measured through a combination of daily quizzes and milestone assessments to ensure consistent learning and identify areas needing additional support.

Assessment	Timing	Format
Daily Quiz / Coding Challenge	Every day (15 min)	Short SQL/PySpark challenge or MCQ to reinforce the day's learning
Mid-Course Practical	Day 8	Build a Silver table, apply optimisation and explain the improvement
Debugging Practical	Day 11	Diagnose and fix intentionally broken pipelines from error logs
Capstone Project	Days 13–15	Team project: end-to-end pipeline with presentation and documentation
Final Hands-On Assessment	Day 15	Individual: fix a broken pipeline and deliver Gold output under time constraint

Additional Evaluation & Accountability Mechanisms

To align with industry expectations, the following will be enforced:

- **Weekly Coding Assessments**
  - SQL + PySpark based evaluations (HackerRank-style problems)
- **Minimum Competency Tracking**
  - Students will be evaluated continuously on performance
- **Assignment Completion Tracking**
  - Daily and weekly submissions are mandatory
- **Corrective Support**
  - Students falling behind will be identified early and guided

 Evaluation will focus not only on correctness but also on:

- Approach
- Efficiency
- Understanding

9. Assumptions

This proposal is prepared based on the following assumptions. Any changes to these assumptions may impact the program scope, delivery approach or timeline.

1. The total batch size is 60 students, trained together as a single cohort.
2. Students have basic Python proficiency (variables, loops, functions, data types). PySpark and Spark concepts will be taught from scratch within the program.
3. Students have been exposed to basic SQL (SELECT, WHERE, JOIN) at an academic level. The program will strengthen and expand SQL skills significantly.
4. The training will be delivered using Azure Databricks as the cloud platform. Storage examples, Auto Loader configurations and cloud awareness content will reference Azure Data Lake Storage (ADLS) and Azure-native services.
5. Databricks workspace access will be provisioned for all 60 students before the program starts. Databricks Community Edition is not sufficient for this scale of training.
6. Each student will have access to a laptop with a modern browser and stable internet connectivity throughout the 15-day program.
7. The program runs Monday to Saturday with Sunday as a holiday, covering 6 hours per day. Exact daily timings (start time, break schedule) will be aligned with the college's regular schedule.

10. Trainer Responsibilities

The trainer will play an active role beyond content delivery:

- Conduct **live coding and debugging sessions**
- Review student submissions and provide feedback
- Identify weak students and support improvement
- Ensure all sessions are **hands-on driven**
- Align delivery with real-world project expectations
- Interact with Capgemini team for feedback and alignment

11. Student Accountability & Participation

To ensure effective learning:

- Attendance is **mandatory for all sessions**
- Daily lab submissions must be completed
- Weekly assessments must be attempted
- Completion of Databricks Academy modules is mandatory
- Active participation in discussions and debugging sessions is expected

 Students are expected to take ownership of their learning and demonstrate consistent effort.